

TCS Interview Master Question Sheet

Complete Technical & HR Preparation: DSA, Core CS, System Design, & Behavioral

Target Roles: TCS Ninja • Digital • Prime

Selection Process: NQT / CodeVita → TR → MR → HR

Format: High-Yield Sheet

1. TCS RECRUITMENT FLOW & EXAM PATTERNS

Interview Rounds Breakdown: Candidates are evaluated across three streams: **Technical (TR)** (core fundamentals + code walkthroughs), **Managerial (MR)** (scenario analysis, project leadership, and real-world trade-offs), and **HR** (adaptability, shifts, relocation, and cultural fit).

Stream / Profile	DSA & Coding Focus	Core CS & Tech Focus	Typical Interview Expectations
TCS Ninja 3.36 - 3.6 LPA	Basic logic, Strings, Arrays, Recursion, Number theory, Pattern printing, Linear Search, Sorting.	OOPs concepts, basic DBMS (CRUD queries), basics of C/C++/Java/Python, SDLC life cycle.	Write bug-free syntax on paper/notepad; explain college final-year project clearly.
TCS Digital 7.0 - 7.5 LPA	Binary Trees, BST, Linked Lists, Stacks, Queues, Sliding Window, Greedy, Dynamic Programming basics.	Normalization, Joins, Indexes, OS Process scheduling, Deadlocks, Threading, REST APIs, Cloud basics.	Time & Space complexity proof, edge-case optimization, system architecture of projects.
TCS Prime 9.0 - 11.5 LPA	Graph traversals (BFS/DFS, Dijkstra), Advanced DP, Bit Manipulation, Trie, Trie + HashMaps.	Microservices, Docker/K8s fundamentals, System scaling, CAP theorem, NoSQL vs SQL trade-offs.	Live coding complex algorithmic logic, in-depth architectural defense, competitive programming caliber.

2. DSA: MOST ASKED INTERVIEW QUESTIONS

Categorized by topic and frequently verified across actual TCS NQT and face-to-face Technical rounds.

A. Arrays & Mathematical Logic

1. Kadane's Algorithm (Max Subarray Sum)

HIGH FREQUENCY

Find contiguous subarray with maximum sum. Discuss handling of all-negative elements. Target: $O(N)$ time, $O(1)$ space.

2. Trapping Rainwater & Two Pointers

Compute volume of trapped water between elevation heights. Compare prefix/suffix array method vs two-pointer $O(1)$ memory method.

3. Dutch National Flag (Sort 0s, 1s, 2s)

Sort in single pass without using standard counting arrays. State pointer logic: `low`, `mid`, `high`.

B. Strings & Hashing

1. Longest Substring Without Repeating Characters **DIGITAL**

Implement sliding window technique with dynamic window expansion and hash map indexing.

2. Valid Anagram & Group Anagrams

Evaluate frequency array approach vs character sorting. Explain hashing tuple/sorted keys.

3. String Compression & Run-Length Encoding

Compress "aaabbccccc" to "a3b2c4". In-place mutation considerations in C/C++.

4. Longest Palindromic Substring

Explain expand around center method $O(N^2)$ vs Dynamic Programming table vs Manacher's Algorithm.

4. TCS Signature: Custom Base & Series Operations

E.g., Prime numbers up to N, Armstrong numbers, Fibonacci via DP, checking if a number is Harshad / Automorphic / Strong.

C. Linked Lists & Stacks / Queues

1. Reverse a Linked List (Iterative & Recursive)

Explain 3-pointer technique (prev , curr , next) and recursion stack unwinding.

2. Detect & Remove Loop (Floyd's Cycle Algorithm)

Prove why slow and fast pointers meet inside the loop and mathematical derivation to find entry point.

3. Next Greater Element & Stock Span Problem

Use monotonic stack to achieve optimal $O(N)$ instead of brute-force $O(N^2)$.

4. Implement Queue using Stacks / Stack using Queues

Explain amortized analysis for push-friendly vs pop-friendly variations.

D. Trees, Graphs & Dynamic Programming

1. Lowest Common Ancestor (LCA) in Binary Tree & BST

Contrast property exploitation in BST vs general BT traversal logic.

2. Tree Traversals & Diameter of Binary Tree

Inorder, Preorder, Postorder (iterative using stack), Level Order (BFS), Boundary traversal, Diameter in $O(N)$.

3. 0/1 Knapsack & Coin Change Problem

DIGITAL / PRIME

Write recurrence relation, memoization table, and space-optimized 1D array DP approach.

4. Graph Traversals & Cycle Detection

BFS & DFS implementations; cycle detection in directed graphs (recursion stack/Kahn's algo) & undirected graphs (parent pointer/Disjoint Set).

3. CORE COMPUTER SCIENCE ESSENTIALS (TR ROUND)

Subject	High-Probability Conceptual & Scenario Questions
OOP Concepts (Java / C++ / Python)	- Explain the 4 pillars (Encapsulation, Abstraction, Inheritance, Polymorphism) with real-world examples (e.g., Car, Bank Account). - Method Overloading vs Overriding: compile-time vs runtime binding, virtual table (vtable) mechanics in C++. - Abstract Class vs Interface: when to choose which? Java 8 default & static interface methods. - Diamond problem in multiple inheritance and how interfaces / C++ virtual base classes resolve it. - Memory management: Stack vs Heap, Garbage Collection algorithms (generational collection, Mark & Sweep).
DBMS & SQL	- ACID properties detailed: Atomicity, Consistency, Isolation levels (Dirty Read, Non-repeatable read, Phantom read), Durability. - Normalization: 1NF, 2NF, 3NF, BCNF with functional dependencies and decomposition scenarios. - SQL Queries: Find Nth highest salary (using subquery, <code>DENSE_RANK()</code>, <code>LIMIT/OFFSET</code>), Joins (Inner, Left, Right, Full Outer, Cross). - Indexing: Clustered vs Non-Clustered index, B-Trees vs Hash Indexing, index fragmentation. - SQL vs NoSQL: Schema flexibility, horizontal scaling (sharding) vs vertical scaling.

Subject	High-Probability Conceptual & Scenario Questions
Operating Systems	• Process vs Thread: Process Control Block (PCB), context switching overhead, multithreading advantages. • CPU Scheduling algorithms: FCFS, SJF, Round Robin (time quantum trade-offs), Priority scheduling. • Deadlocks: 4 Coffman conditions, Banker's algorithm for avoidance, detection & recovery strategies. • Memory Management: Virtual Memory, Paging, Page Faults, Thrashing, Page replacement policies (FIFO, LRU).
Computer Networks	• OSI 7 Layers vs TCP/IP model: Exact functions and protocols at each layer (HTTP, TCP, IP, ARP, MAC). • TCP vs UDP: 3-way handshake, flow control (sliding window), congestion control, when UDP is preferred. • DNS resolution: Step-by-step trace of typing https://www.google.com in a web browser. • HTTP vs HTTPS: SSL/TLS handshake, symmetric vs asymmetric encryption certificates.

4. PROJECT DEFENSE & MANAGERIAL (MR) SCENARIOS

Project Deep-Dive Architecture

- **Role & Contribution:** Clearly state what *you* developed vs teammates (avoid saying "we did everything").
- **Tech Stack Justification:** Why did you choose React/Node or Django/PostgreSQL over alternatives?
- **Database Schema:** Be ready to draw the Entity-Relationship (ER) diagram and explain primary/foreign keys.
- **Challenges & Bug Fixes:** Detail one significant technical hurdle encountered and the systematic debugging method used.
- **Security & Scalability:** How do you authenticate users (JWT, OAuth) and handle simultaneous traffic?

Managerial Scenarios (MR Round)

- **Tight Deadlines:** What would you do if your project requirement changes two days before deployment?
- **Team Conflict:** How do you resolve a technical disagreement with a team member regarding design patterns?
- **Legacy Systems:** Are you willing to work on support/maintenance or legacy codebases instead of new tech?
- **Client Communication:** How would you explain a major production outage to a non-technical stakeholder?
- **Learning Curve:** Describe a time you picked up an entirely new framework or language in under a week.

5. HR ROUND: BEHAVIORAL & MANDATORY POLICIES

Crucial HR Tip: TCS heavily values integrity, adaptability, and flexibility. Any hesitation regarding relocation, 24x7 shifts, or training bonds can adversely affect your candidatures.

Classic HR Question	Ideal Strategy & Answering Framework
Tell me about yourself.	Structure chronologically: Education → Technical Strengths & Hands-on Projects → Extracurricular/Leadership → Why TCS. Keep under 90 seconds. Avoid repeating personal details already in resume.
Why Tata Consultancy Services (TCS)?	Highlight Tata Group ethics, massive global footprint, investment in upskilling (TCS Elevate, Wings1 programs), job security, and diverse enterprise client portfolio.
Are you ready to relocate and work in rotational / night shifts?	Always answer with an emphatic "Yes". State that learning and business demands take priority early in your engineering career.

Classic HR Question	Ideal Strategy & Answering Framework
Where do you see yourself in 5 years?	Frame around technical mastery: transitioning from a software engineer to a full-stack technical lead or cloud architect delivering large-scale enterprise solutions within TCS.
What are your strengths and weaknesses?	Strength: Rapid learning, structured problem solving, team collaboration. Weakness: Detail-oriented to a fault (and explain how you manage it with time-boxing and prioritization).
Do you have any questions for us?	Never say "No". Ask: "What tech stacks are currently dominating new digital projects in your unit?" or "What advice would you give a fresher joining this year?"

6. LAST-MINUTE INTERVIEW CHECKLIST (DAY BEFORE)

Technical Checklist:

- Master time complexities of standard algorithms (Sorting, Hash Map, Heaps).
- Be able to write syntactically correct code on plain paper or blank Notepad.
- Memorize 2 SQL queries: Nth highest salary and 2nd table Join with aggregation.
- Review all resume projects and prepare 2-minute project elevator pitch.

Etiquette & Documentation:

- Keep government ID (Aadhaar / PAN / Passport) and college ID ready.
- Formal dress code (plain collared shirt/blazer, groomed appearance).
- Quiet environment with stable internet and functional mic/webcam.
- Keep a notebook and pen beside you for dry-running algorithmic traces.